Activity 1

(i) Take a graph sheet.

(ii) Consider a triangle whose base is the line joining the points (0,0) and (6,0)

(iii) Take the third vertex as (1,1), (2,2), (3,3), (4,4), (5,5) and find their areas. Fill in the details given.

(iv) Do you see any pattern with A_1, A_2, A_3, A_4, A_5 ? If so mention it.

(v) Repeat the same process by taking third vertex in step (iii) as $(1,2)$, $(2,4)$, $(3,8)$, $(4,16)$, $(5,32)$.

(vi) Fill the table with these new vertices.

(vii) What pattern do you observe now with A_1, A_2, A_3, A_4, A_5?

Third vertex	Area of Triangle
$(1,1)$	$A_1 =$
$(2,2)$	$A_2 =$
$(3,3)$	$A_3 =$
$(4,4)$	$A_4 =$
$(5,5)$	$A_5 =$

Third vertex	Area of Triangle
$(1,2)$	$A_1 =$
$(2,4)$	$A_2 =$
$(3,8)$	$A_3 =$
$(4,16)$	$A_4 =$
$(5,32)$	$A_5 =$

Activity 2

Find the area of the shaded region

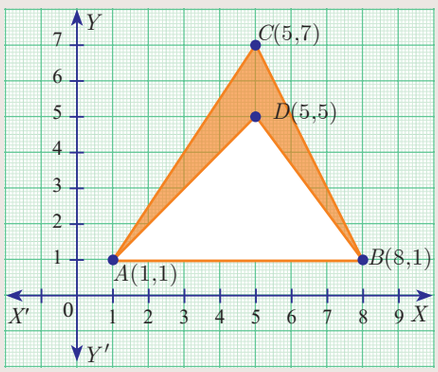


Fig. 5.15

DO YOU KNOW?

Two French mathematicians Rene Descartes and Pierre-de-Fermat were the first to conceive the idea of modern coordinate geometry by 1630s.

Exercise 5.1

1. Find the area of the triangle formed by the points
   (i) (1,–1), (–4, 6) and (–3, –5) (ii) (–10, –4), (–8, –1) and (–3, –5)

2. Determine whether the sets of points are collinear?
   (i) $\left(-\frac{1}{2}, 3\right)$, (–5, 6) and (–8, 8) (ii) $(a, b+c)$, $(b, c+a)$ and $(c, a+b)$

3. Vertices of given triangles are taken in order and their areas are provided aside. In each case, find the value of 'p '.

S.No.	Vertices	Area (sq.units)
(i)	$(0, 0)$, $(p, 8)$, $(6, 2)$	20
(ii)	(p, p), $(5, 6)$, $(5, -2)$	32

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